Broker API Integration Requirements Specification

Abaci Investments Limited

Technical specification for API connectivity between Abaci Investments (Merchant Bank / Custodian / Portfolio Manager) and executing broker(s) for the Bangladesh capital market.

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Intended Audience: Broker CTO / Tech Team / API Integration Team

Regulatory Context: BSEC Licensed Merchant Bank Operations

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1. Executive Summary

Abaci Investments Limited operates as a **BSEC-licensed Merchant Bank** providing custodial and portfolio management services for clients trading on the Dhaka Stock Exchange (DSE) and Chittagong Stock Exchange (CSE). As a custodian and portfolio manager, Abaci does not execute trades directly — trade execution is the responsibility of our partner broker(s).

This document specifies the **API integration requirements** for connecting Abaci's back-office system with the broker's order management and execution infrastructure. The integration enables:

- **Order Routing:** Abaci sends buy/sell orders on behalf of managed clients to the broker for execution on DSE/CSE
- **Execution Reports:** Broker sends real-time/batch execution confirmations back to Abaci
- **Position & Settlement Data:** Broker provides daily position statements and settlement reports for reconciliation
- **Reconciliation:** Automated matching of Abaci's records (system of record) against broker's execution data

> **Key Principle:** Abaci's back-office is the **primary system of record** for all client portfolios, holdings, and fund accounting. Broker data is consumed for execution confirmation and reconciliation purposes. Any discrepancy is flagged and resolved through the defined process.

2. Business Context & Relationship Model

2.1 Abaci's Role (Merchant Bank / Custodian / Portfolio Manager)

- Manages client portfolios and makes investment decisions
- Holds securities in custody via CDBL Depository Participant membership
- Maintains the master record of client holdings, cash positions, and NAV
- Initiates orders based on portfolio strategy and client mandates
- Performs daily NAV calculation and client fund accounting
- Handles corporate actions (dividends, rights, bonus) for custodial clients
- Generates regulatory reports (BSEC, DSE, CDBL)

2.2 Broker's Role (Trade Execution)

- Receives orders from Abaci and routes to DSE/CSE for execution
- Manages exchange connectivity, order book, and execution quality
- Sends execution reports (fills, partial fills, rejections) back to Abaci
- Provides daily position/trade statements for reconciliation
- Handles exchange-level settlement (T+2) with clearing house
- Calculates and applies brokerage fees, exchange fees, regulatory charges

2.3 Data Flow Architecture

```
┌──────────────────────┐          ┌──────────────────────┐
│   ABACI INVESTMENTS  │          │   BROKER PARTNER     │
│   (Merchant Bank)    │          │   (Trade Execution)  │
│                      │          │                      │
│  Portfolio Manager   │──────────│  Order Management    │
│  ├─ Order Decision   │   API    │  ├─ Order Validation  │
│  ├─ Send Order ──────│──────────>│  ├─ Route to DSE/CSE │
│  │                   │          │  ├─ Execute          │
│  ├─ Receive Fill  <──│<─────────│──├─ Send Fill Report  │
│  ├─ Update Holdings  │          │  │                   │
│  ├─ Update Ledger    │          │  └─ Settlement       │
│  │                   │          │                      │
│  Reconciliation      │          │  Daily Reports       │
│  ├─ Pull Positions <─│<─────────│──├─ Position Stmt    │
│  ├─ Pull Trades   <──│<─────────│──├─ Trade Stmt       │
│  ├─ Match & Flag     │          │  └─ Settlement Stmt  │
│  └─ Resolve          │          │                      │
└──────────────────────┘          └──────────────────────┘
```

3. API Integration Requirements

3.1 Order Submission API REQUIRED PHASE 1

Abaci submits orders to the broker for execution. Each order represents a portfolio manager's investment decision for a specific client account.

FIELD	TYPE	REQUIRED	DESCRIPTION
`order_id`	String	Yes	Abaci's unique order reference (UUID)
`client_code`	String	Yes	Client's BO account number or broker client code
`portfolio_id`	String	Yes	Abaci's internal portfolio reference
`symbol`	String	Yes	DSE/CSE trading code (e.g., "BRACBANK")
`isin`	String	No	ISIN code if available
`exchange`	Enum	Yes	`DSE` | `CSE`
`side`	Enum	Yes	`BUY` | `SELL`
`order_type`	Enum	Yes	`MARKET` | `LIMIT`
`quantity`	Integer	Yes	Number of shares
`limit_price`	Decimal	Conditional	Required for LIMIT orders. BDT per share
`time_in_force`	Enum	No	`DAY` (default) | `IOC` | `GTC`
`board`	Enum	No	`PUBLIC` (default) | `BLOCK` | `SPOT`
`notes`	String	No	Free-text instructions
`timestamp`	ISO 8601	Yes	Order submission time (UTC+6)

Expected Response:

```
{
  "status": "accepted",          // accepted | rejected | pending
  "broker_order_id": "BRK-2026-00451", // Broker's reference
  "abaci_order_id": "uuid-here",      // Echo back
  "received_at": "2026-04-17T10:30:00+06:00",
  "rejection_reason": null            // Populated if rejected
}
```

3.2 Execution Report API REQUIRED PHASE 1

Broker sends execution reports (fills) to Abaci. This can be via:

- **Push (Webhook):** Broker POSTs to Abaci's endpoint on each fill — preferred for real-time
- **Pull (Polling):** Abaci queries broker's API periodically — acceptable for batch processing
- **File (Batch):** Broker provides end-of-day trade file — minimum acceptable standard

FIELD	TYPE	REQUIRED	DESCRIPTION
`exec_id`	String	Yes	Broker's unique execution ID (for dedup)
`broker_order_id`	String	Yes	Broker's order reference
`abaci_order_id`	String	Yes	Abaci's original order reference
`client_code`	String	Yes	Client BO / broker code
`symbol`	String	Yes	Trading symbol
`isin`	String	No	ISIN code
`exchange`	Enum	Yes	`DSE` | `CSE`
`side`	Enum	Yes	`BUY` | `SELL`
`exec_type`	Enum	Yes	`FILL` | `PARTIAL_FILL` | `CANCEL` | `REJECT`
`exec_qty`	Integer	Yes	Executed quantity (this fill)
`exec_price`	Decimal	Yes	Execution price (BDT)
`gross_value`	Decimal	Yes	exec_qty × exec_price
`commission`	Decimal	Yes	Brokerage commission amount
`exchange_fee`	Decimal	Yes	Exchange/Laga fee
`cdbl_fee`	Decimal	Yes	CDBL settlement fee
`ait`	Decimal	Yes	Advance Income Tax
`net_value`	Decimal	Yes	Net amount (gross ± all fees)
`trade_date`	Date	Yes	Trade execution date
`settlement_date`	Date	Yes	Expected settlement date (T+2/T+3)
`exec_time`	ISO 8601	Yes	Exact execution timestamp
`cum_qty`	Integer	No	Cumulative filled quantity for this order
`leaves_qty`	Integer	No	Remaining unfilled quantity
`category`	String	No	Stock category (A/B/Z/N/G) — for settlement calc

3.3 Order Status & Management API REQUIRED PHASE 1

ENDPOINT	METHOD	DESCRIPTION
`GET /orders/{abaci_order_id}`	GET	Query status of a specific order
`GET /orders?date={date}&client={code}`	GET	List all orders for a date/client
`DELETE /orders/{abaci_order_id}`	DELETE	Cancel a pending order
`PUT /orders/{abaci_order_id}`	PUT	Modify pending order (qty/price)

Order Status Values:

STATUS	DESCRIPTION
`PENDING`	Order received, awaiting routing to exchange
`OPEN`	Order placed on exchange, awaiting execution
`PARTIALLY_FILLED`	Some quantity executed, balance still open
`FILLED`	Fully executed
`CANCELLED`	Cancelled by Abaci or auto-cancelled (end of day)
`REJECTED`	Rejected by broker or exchange (with reason)
`EXPIRED`	Order expired (time in force exceeded)

3.4 Portfolio & Position Query API REQUIRED PHASE 1

Abaci needs to query the broker's view of client positions for reconciliation. This should reflect the broker's record of holdings for each client.

ENDPOINT	METHOD	DESCRIPTION
`GET /positions/{client_code}`	GET	Current positions for a client
`GET /positions?date={date}`	GET	All client positions as of a date
`GET /positions/{client_code}/{symbol}`	GET	Specific holding detail

Position Record Fields:

FIELD	TYPE	DESCRIPTION
`client_code`	String	Client identifier
`symbol`	String	Trading symbol
`isin`	String	ISIN code
`total_qty`	Integer	Total quantity held
`saleable_qty`	Integer	Qty available for sell (excl. unsettled)
`avg_cost`	Decimal	Weighted average cost per share
`market_price`	Decimal	Current/last market price
`market_value`	Decimal	total_qty × market_price
`unsettled_buy_qty`	Integer	Pending buy settlement
`unsettled_sell_qty`	Integer	Pending sell settlement
`as_of_date`	Date	Position date

3.5 Settlement & Reconciliation API REQUIRED PHASE 2

ENDPOINT	METHOD	DESCRIPTION
`GET /settlements?date={date}`	GET	Settlements due/completed on a date
`GET /settlements/{client_code}?from={date}&to={date}`	GET	Client settlement history
`GET /reconciliation/daily?date={date}`	GET	Full daily recon report (all clients)
`GET /statements/{client_code}?date={date}`	GET	Client's daily statement (trades + balance)

Settlement Record Fields:

FIELD	TYPE	DESCRIPTION
`settlement_id`	String	Unique settlement reference
`exec_id`	String	Related execution ID
`client_code`	String	Client identifier
`symbol`	String	Security symbol
`side`	Enum	BUY | SELL
`quantity`	Integer	Settlement quantity
`amount`	Decimal	Net settlement amount (BDT)
`trade_date`	Date	Original trade date
`settlement_date`	Date	Settlement due date
`status`	Enum	PENDING | SETTLED | FAILED
`settled_date`	Date	Actual settlement date (if settled)

3.6 Market Data Feed OPTIONAL PHASE 2

If the broker provides market data via API, Abaci would consume:

- End-of-day closing prices for NAV calculation
- Intraday LTP for real-time portfolio valuation
- Corporate action announcements (dividends, rights, bonus)

> **Note:** Market data is optional from the broker. Abaci currently sources market data from DSE directly. However, if the broker can provide this, it simplifies the data pipeline and ensures price consistency for reconciliation.

4. Data Schemas & Field Specifications

4.1 Daily Reconciliation File Format

If real-time API is not available initially, Abaci can accept daily batch files in the following formats (in order of preference):

PRIORITY	FORMAT	DESCRIPTION
1	JSON API	RESTful API responses as specified above
2	CSV/Excel	Structured flat file with defined column headers
3	XML	DSE standard XML format (as used for trade files)
4	FIX Protocol	Financial Information eXchange standard messages

Required Daily Files (if batch mode):

File 1: Daily Trade Report — `trades_YYYYMMDD.csv`

```
exec_id, client_code, symbol, isin, exchange, side, quantity, price,
gross_value, commission, exchange_fee, cdbl_fee, ait, net_value,
trade_date, settlement_date, trade_time, category, board
```

File 2: Daily Position Statement — `positions_YYYYMMDD.csv`

```
client_code, symbol, isin, total_qty, saleable_qty, avg_cost,
market_price, market_value, unsettled_buy_qty, unsettled_sell_qty
```

File 3: Settlement Report — `settlements_YYYYMMDD.csv`

```
settlement_id, exec_id, client_code, symbol, side, quantity,
amount, trade_date, settlement_date, status
```

File 4: Cash Balance Report — `cash_YYYYMMDD.csv`

```
client_code, ledger_balance, available_balance, receivables,
payables, margin_used, equity
```

5. Technical Requirements

5.1 Authentication & Security

REQUIREMENT	SPECIFICATION
Authentication	API Key + Secret (HMAC-SHA256) or OAuth 2.0 Client Credentials
Transport	HTTPS/TLS 1.2+ mandatory. No plaintext HTTP.
IP Whitelisting	Abaci will provide static IP(s) for whitelisting
Request Signing	HMAC signature on request body to prevent tampering
Idempotency	All POST endpoints must support idempotency keys to prevent duplicate orders
Encryption	Sensitive fields (NID, account numbers) encrypted at rest

5.2 Transport & Protocol

REQUIREMENT	SPECIFICATION
Protocol	REST over HTTPS (primary), WebSocket for real-time fills (optional)
Data Format	JSON (Content-Type: application/json)
Character Encoding	UTF-8
Date Format	ISO 8601 (YYYY-MM-DD for dates, full ISO for timestamps)
Currency	BDT (all monetary values in Bangladeshi Taka)
Decimal Precision	Prices: 2 decimal places. Quantities: integers. Fees: 2 decimal places.
Pagination	Cursor-based or offset/limit for list endpoints

5.3 Rate Limits & SLA

METRIC	REQUIREMENT
Order Submission	≤ 500ms response time (95th percentile)
Execution Reports	≤ 2 second delivery latency (webhook) or ≤ 5 min polling interval
Position Queries	≤ 2 second response time
API Availability	99.5% uptime during market hours (Sun-Thu 10:00-14:30 BST)
Rate Limit	Minimum 100 requests/minute per API key
Daily File Delivery	By 17:00 BST on trade date (if batch mode)
Support Response	≤ 30 minutes during market hours for P1 issues

5.4 Error Handling

All API responses must follow a consistent error format:

```
{
  "success": false,
  "error": {
    "code": "INSUFFICIENT_BALANCE",
    "message": "Client does not have sufficient funds for this order",
    "details": {
      "required": 150000.00,
      "available": 120000.00
    }
  },
  "request_id": "req-uuid",
  "timestamp": "2026-04-17T10:30:00+06:00"
}
```

Standard Error Codes:

CODE	HTTP	DESCRIPTION
`INVALID_SYMBOL`	400	Symbol not found or not tradeable
`INSUFFICIENT_BALANCE`	400	Client has insufficient funds
`INSUFFICIENT_HOLDINGS`	400	Client doesn't hold enough shares to sell
`MARKET_CLOSED`	400	Order submitted outside market hours
`PRICE_OUT_OF_RANGE`	400	Limit price outside circuit breaker range
`DUPLICATE_ORDER`	409	Order with same idempotency key already exists
`ORDER_NOT_FOUND`	404	Order ID not found
`UNAUTHORIZED`	401	Invalid API key or signature
`RATE_LIMIT`	429	Too many requests
`INTERNAL_ERROR`	500	Broker system error

5.5 Environments

ENVIRONMENT	PURPOSE	DATA
Sandbox / UAT	Integration testing with mock data	Simulated orders, test symbols
Production	Live trading	Real DSE/CSE orders

> **Requirement:** A sandbox environment with simulated order matching is required before production go-live. Abaci will not route live orders until end-to-end testing is validated in sandbox.

6. Reconciliation Process

6.1 Daily Reconciliation Workflow

Reconciliation runs daily after market close and compares Abaci's records (system of record) against broker's data:

STEP	TIME	ACTION	OWNER
1	14:30 BST	Market closes. Broker finalizes trade data.	Broker
2	15:00 BST	Abaci pulls/receives execution reports for the day	Abaci
3	15:30 BST	Abaci pulls broker's position statement for all clients	Abaci
4	16:00 BST	Automated reconciliation runs: match Abaci holdings vs broker positions	Abaci
5	16:00 BST	Discrepancy report generated (if any mismatches)	Abaci
6	16:30 BST	Discrepancies communicated to broker for investigation	Both
7	17:00 BST	NAV calculation finalized using reconciled data	Abaci

6.2 Reconciliation Points

CHECK	ABACI FIELD	BROKER FIELD	TOLERANCE
Trade Count	portfolio_transactions count	Daily trade report count	Exact match
Execution Price	transaction.price	exec_report.exec_price	Exact match
Executed Qty	transaction.quantity	exec_report.exec_qty	Exact match
Holdings Qty	portfolio_holdings.quantity	position.total_qty	Exact match
Avg Cost	portfolio_holdings.avg_cost	position.avg_cost	BDT 0.10 (rounding)
Net Value	transaction.net_value	exec_report.net_value	BDT 1.00 (rounding)
Cash Balance	fund_accounts.cash_balance	cash_report.ledger_balance	BDT 10.00
Settlement Status	settlement_tracking.status	settlement.status	Exact match

6.3 Settlement Calendar (Bangladesh)

CATEGORY	BUY SETTLEMENT	SELL SETTLEMENT
Category A, B, G, N	T+2 business days	T+2 business days
Category Z	T+3 business days	T+2 business days
Spot / Compulsory Spot	T+1 business day	T+0 (same day)
Block Trade	T+2 business days	T+2 business days

Business Days: Sunday through Thursday. Excludes Bangladesh public holidays and exchange-declared holidays.

7. Phased Implementation Plan

PHASE	DURATION	SCOPE	DELIVERABLES
Phase 1A Foundation	2-3 weeks	• Broker provides API documentation • API key provisioning • Sandbox environment setup • Client code mapping (Abaci ↔ Broker)	API credentials, endpoint URLs, sandbox access, client code mapping table
Phase 1B Order Flow	3-4 weeks	• Order submission integration • Execution report consumption • Order status tracking • End-to-end test with sandbox	Working order flow in sandbox, test results, error handling validated
Phase 2A Reconciliation	2-3 weeks	• Position statement API integration • Daily reconciliation automation • Discrepancy flagging & reporting • Settlement tracking	Automated daily recon, discrepancy dashboard, settlement calendar
Phase 2B Go-Live	1-2 weeks	• Production credentials • Parallel run (manual + API for 5 days) • Production cutover • Monitoring & alerting	Production API live, monitoring dashboard, runbook
Phase 3 Enhancement	Ongoing	• Real-time WebSocket fills • Market data feed • Multi-broker support • Corporate action automation	Enhanced real-time capabilities, multi-broker routing

8. Appendices

A. Fee Schedule (Current BSEC Standard)

FEE COMPONENT	RATE	BASIS	NOTES
Brokerage Commission	Max 0.40%	Trade value	Negotiable between Abaci & Broker
Exchange Fee (Laga)	0.03%	Trade value	0.0125% for block trades ≥ BDT 5M
CDBL Fee	0.0175%	Trade value	Minimum BDT 5 per transaction
AIT (Advance Income Tax)	0.05%	Trade value	Changing to 0.03% effective July 2026

B. BSEC Regulatory References

- BSEC (Alternative Trading System) Rules, 2021
- BSEC (Stock Dealer, Stock Broker and Authorized Representative) Rules, 2000 (amended)
- BSEC Guidelines on Electronic Know-Your-Customer (e-KYC)
- BSEC (Margin) Rules, 2025
- CDBL Operational Procedures
- DSE & CSE Trading Rules and Settlement Procedures

C. Contact Information

ROLE	CONTACT	RESPONSIBILITY
Technical Lead	tech@abaci-investments.com	API integration, testing, go-live
Operations Lead	ops@abaci-investments.com	Client mapping, reconciliation, settlement
Compliance	compliance@abaci-investments.com	Regulatory requirements, audit trail

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